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QUARTERLY EXAMINATION - 2025

CLASS : 12 **PHYSICS** Reg.No :

TIME : 3.00 Hrs. MARKS : 70

PART – I

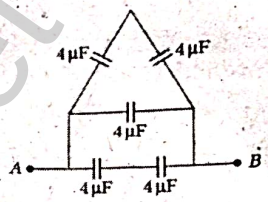
$15 \times 1 = 15$

Note: **(i) Answer all the questions**
(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer

1. Two coherent monochromatic light beams of intensities *I* and *4I* are superposed. The maximum and minimum possible intensities in the resulting beam are
   a) 5I and I b) 5I and 3I c) 9I and I d) 9I and 3I

2. An electric field $\vec{E} = 10\,x\,\hat{\imath}$ exists in a certain region of space. Then the potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A is the potential at x = 2 m is:
   a) 10 V b) - 20 V c) + 20 V d) - 10 V

3. Equivalent capacitance between A and B is
   a) $8\mu F$ b) $6\mu F$
   c) $26\mu F$ d) $\frac{10}{3}\mu F$



4. A thin insulated wire forms a plane spiral of N = 100 tight turns carrying a current I = 8 mA (milli ampere). The radii of inside and outside turns are a = 50 mm and b = 100 mm respectively. The magnetic induction at the center of the spiral is
   a) 5 μT b) 7 μT c) 8 μT d) 10 μT

5. A non-conducting charged ring of charge *q*, mass *m* and radius *r* is rotated with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$

6. When a charged particle enters a uniform magnetic field, its kinetic energy
   a) increases b) decreases c) remains constant d) becomes zero

7. If the amplitude of the magnetic field is $3 \times 10^{-6}\ T$, then amplitude of the electric field for a electromagnetic waves is
   a) 100 V m^{-1} b) 300 V m^{-1} c) 600 V m^{-1} d) 900 V m^{-1}

8. In electromagnetic waves, the phase difference between electric and magnetic field vectors $\vec{E}$ and $\vec{B}$ is
   a) zero b) $\frac{\pi}{2}$ c) π d) $\frac{\pi}{4}$

9. A toaster operating at 240 V has a resistance of 120 Ω. The power is
   a) 400 W b) 2 W c) 480 W d) 240 W

10. If a wire is stretched to three time its original length then the specific resistance of the wire will
    a) becomes 9 times b) becomes 81 times c) becomes $\frac{1}{3}$ times d) remains the same

11. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
    a) $\frac{Q}{2}$ b) $\frac{Q}{\sqrt{3}}$ c) $\frac{Q}{\sqrt{2}}$ d) Q

12. The flux linked with a coil at any instant t is given by $\emptyset_B = 10t^2 - 50t + 250$. The induced emf at t = 3s is
    a) -190 V b) -10 V c) 10 V d) 190 V

13. With increase in frequency of an AC supply the impedance of an LCR series circuit
    a) remains constant b) increases
    c) decreases d) decreases first, becomes minimum and then increases

14. For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is,
    a) 30° b) 45° c) 60° d) 90°

15. A plan convex lens is made of glass of refractive index 1.5. The radius of curvature of its convex surface is R. Its focal length is
    a) $\frac{R}{2}$ b) R c) $2R$ d) $\frac{3}{2}R$

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PART – II $6 \times 2 = 12$

Answer any six questions. Question no. 20 is compulsory:

16. What are the properties of an equipotential surface?
17. Define resistivity.
18. What is Seebeck effect?
19. Write the limitations of cyclotron.
20. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. Calculate the magnetic dipole moment of the coil.
21. State Huygens' Principle.
22. What is displacement current?
23. Define critical angle.
24. A circular antenna of area 3 m^2 is installed at a place in Madurai. The plane of the area of antenna is inclined at 47° with the direction of Earth's magnetic field. If the magnitude of Earth's field at that place is 4.1×10^{-5} T. Find the magnetic flux linked with the antenna.

PART – III $6 \times 3 = 18$

Answer any six questions. Question no. 29 is compulsory:

25. Derive the expression for resultant capacitance of capacitors connected in series.
26. Explain the determination of the internal resistance of a cell using voltmeter.
27. Discuss the conversion of galvanometer into voltmeter.
28. Explain the process of electrostatic induction.
29. A copper wire of cross-sectional area 0.5 mm^2 carries a current of 0.2 A. If the free electron density of copper is 8.4×10^{28} m^{-3} then compute the drift velocity of free electrons.
30. What are the advantages and disadvantages of alternating current system over direct current system?
31. What is optical path? Obtain the equation for optical path of a medium of thickness d and refractive index n.
32. Write down the properties of electromagnetic waves.
33. What is the focal length of the combination if a lens of focal length -70 cm is brought in contact with a lens of focal length 50 cm? What is the power of the combination?

PART – IV $5 \times 5 = 25$

Answer all the questions:

34. a) Derive an expression for electrostatic potential due to an electric dipole.

(Or)

b) Find out the phase relationship between voltage and current in a pure inductive circuit.

35. a) Explain the determination of unknown resistance using meter bridge.

(Or)

b) Obtain the equation for bandwidth in Young's double slit experiment.

36. a) Obtain a relation for the magnetic induction at a point due to an infinitely long straight conductor carrying current

(Or)

b) What is absorption spectra? Explain their types.

37. a) Explain the construction and working of transformer

(Or)

b) Derive the mirror equation and the equation for lateral magnification.

38. a) (i) State Gauss law.
(ii) Calculate the electric flux through the rectangle of sides 5 cm and 10 cm kept in the region of uniform electric field 100 N C^{-1}. The angle θ (angle between electric field and area vector) is 60°. If θ becomes zero, what is the electric flux?

(Or)

b) (i) Define current sensitivity and voltage sensitivity of a galvanometer.
(ii) The coil of a moving coil galvanometer has 5 turns and each turn has an effective area of 2×10^{-2} m^2. It is suspended in a magnetic field whose strength is 4×10^{-2} $Wb\,m^{-2}$. If the torsional constant K of the suspension fibre is 4×10^{-9} $N\,m\,deg^{-1}$. (a) Find its current sensitivity in division per micro - ampere. (b) Calculate the voltage sensitivity of the galvanometer for it to have full scale deflection of 50 divisions for 25 mV.

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